Floating Wigner Crystal and Periodic Jellium Configurations

Asbjørn Bækgaard Lauritsen
asbjornbaekgaard.lauritsen@ist.ac.at

IST AUSTRIA
Institute of Science and Technology

Current Topics in Mathematical Physics
Zürich
2021-07-20

Definition (Jellium)

Domain $\Omega_N \subset \mathbb{R}^2$ of size $|\Omega_N| = N$ sufficiently regular. Jellium energy is

$$\begin{aligned}\mathcal{E}_{\mathrm{Jel}} = &\sum_{j<k} -\log|x_j - x_k| \\ &+ \sum_{j=1}^{N} \int_{\Omega_N} \log|x_j - y| \,\mathrm{d}y \\ &- \frac{1}{2} \iint_{\Omega_N \times \Omega_N} \log|x - y| \,\mathrm{d}x \,\mathrm{d}y.\end{aligned}$$

Thermodynamic limit exists (Sari–Merlini 1976), $e_{\mathrm{Jel}} \geq -0.66118$.

Definition (Jellium)

Domain $\Omega_N \subset \mathbb{R}^2$ of size $|\Omega_N| = N$ sufficiently regular. Jellium energy is

$$\mathcal{E}_{\mathrm{Jel}} = \sum_{j<k} -\log|x_j - x_k| + \sum_{j=1}^{N} \int_{\Omega_N} \log|x_j - y| \,\mathrm{d}y - \frac{1}{2} \iint_{\Omega_N \times \Omega_N} \log|x-y| \,\mathrm{d}x \,\mathrm{d}y.$$

Thermodynamic limit exists (Sari–Merlini 1976), $e_{\mathrm{Jel}} \geq -0.66118$.

Definition (UEG)

The indirect energy of a charge distribution ρ is

$$\mathcal{E}_{\mathrm{Ind}} = \inf_{\mathbb{P}:\rho_{\mathbb{P}}=\rho} \left[\int \sum_{j<k} -\log|x_j - x_k| \,\mathrm{d}\mathbb{P} + \frac{1}{2} \iint \log|x-y| \rho(x)\rho(y) \,\mathrm{d}x \,\mathrm{d}y \right],$$

where $\rho_{\mathbb{P}}$ is the one-particle density associated to $\mathbb{P}$, i.e. the sum of all the marginals

$$\rho_{\mathbb{P}} = \sum_{j=1}^{N} \int_{\mathbb{R}^{2(N-1)}} \mathrm{d}\mathbb{P}(x_1, \ldots, \hat{x}_j, \ldots, x_N).$$

Thermodynamic limit exists for $\rho = \mathbb{1}_{\Omega_N}$ (Lewin–Lieb–Seiringer 2018)

Connection: Take $\mathbb{P}$ average over translations of crystal configuration

Connection: Take $\mathbb{P}$ average over translations of crystal configuration

Theorem (Lewin–Lieb–Seiringer 2019; Cotar–Petrache 2019)

$$e_{\mathrm{UEG}} = e_{\mathrm{Jel}} \qquad \text{in dimensions } d \geq 3.$$

False in $d = 1$,

$$e_{\mathrm{UEG},d=1} = \frac{1}{6}, \qquad e_{\mathrm{Jel},d=1} = \frac{1}{12}.$$

(Baxter 1963; Kunz 1974; Choquard 1975; Colombo–De Pascale–Di Marino 2015)

Connection: Take $\mathbb{P}$ average over translations of crystal configuration

Theorem (Lewin–Lieb–Seiringer 2019; Cotar–Petrache 2019)

$$e_{\mathrm{UEG}} = e_{\mathrm{Jel}} \qquad \text{in dimensions } d \geq 3.$$

False in $d = 1$,

$$e_{\mathrm{UEG},d=1} = \frac{1}{6}, \qquad e_{\mathrm{Jel},d=1} = \frac{1}{12}.$$

(Baxter 1963; Kunz 1974; Choquard 1975; Colombo–De Pascale–Di Marino 2015)

Theorem (L 2021)

$$e_{\mathrm{UEG}} = e_{\mathrm{Jel}} \qquad \text{in dimensions } d \geq 2.$$

Sketch of proof

$$\mathcal{E}_{\mathrm{Ind}}(\mathbb{P}) = \int \mathcal{E}_{\mathrm{Jel}}(\Omega_N, x_1, \ldots, x_N)\, \mathrm{d}\mathbb{P}(x_1, \ldots, x_N) \geq \min \mathcal{E}_{\mathrm{Jel}}(\Omega_N, x_1, \ldots, x_N).$$

Thus $e_{\mathrm{EUG}} \geq e_{\mathrm{Jel}}$.

For the other inequality:

- Introduce artificial crystal structure - periodic Jellium.
- Good floating crystal trial state - compute its energy.

Naive floating crystal

Consider $n = \ell^2$ particles in cube C_ℓ. Let

$$\Omega_N = \bigcup_{\substack{k \in \mathbb{Z}^2 \\ |k_1|, |k_2| \leq K}} C_\ell + \ell k.$$

Trial state of floating crystal:

$$\mathbb{P} = \frac{1}{\ell^2} \int_{C_\ell} \bigotimes_{\substack{j=1,\dots,n \\ k \in \mathbb{Z}^2 \\ |k_1|, |k_2| \leq K}} \delta_{x_j + \ell k + a} \, \mathrm{d}a, \qquad \rho_{\mathbb{P}} \approx \mathbb{1}_{\Omega_N}$$

Naive floating crystal

Consider $n = \ell^2$ particles in cube C_ℓ. Let

$$\Omega_N = \bigcup_{\substack{k \in \mathbb{Z}^2 \\ |k_1|, |k_2| \leq K}} C_\ell + \ell k.$$

Trial state of floating crystal:

$$\mathbb{P} = \frac{1}{\ell^2} \int_{C_\ell} \bigotimes_{\substack{j=1,\dots,n \\ k \in \mathbb{Z}^2 \\ |k_1|, |k_2| \leq K}} \delta_{x_j + \ell k + a} \, \mathrm{d}a, \qquad \rho_{\mathbb{P}} \approx \mathbb{1}_{\Omega_N}$$

Problem - $O(N \log N)$ shift in energy from charge pileup at boundary. ($O(N)$ in dimension $d \neq 2$)

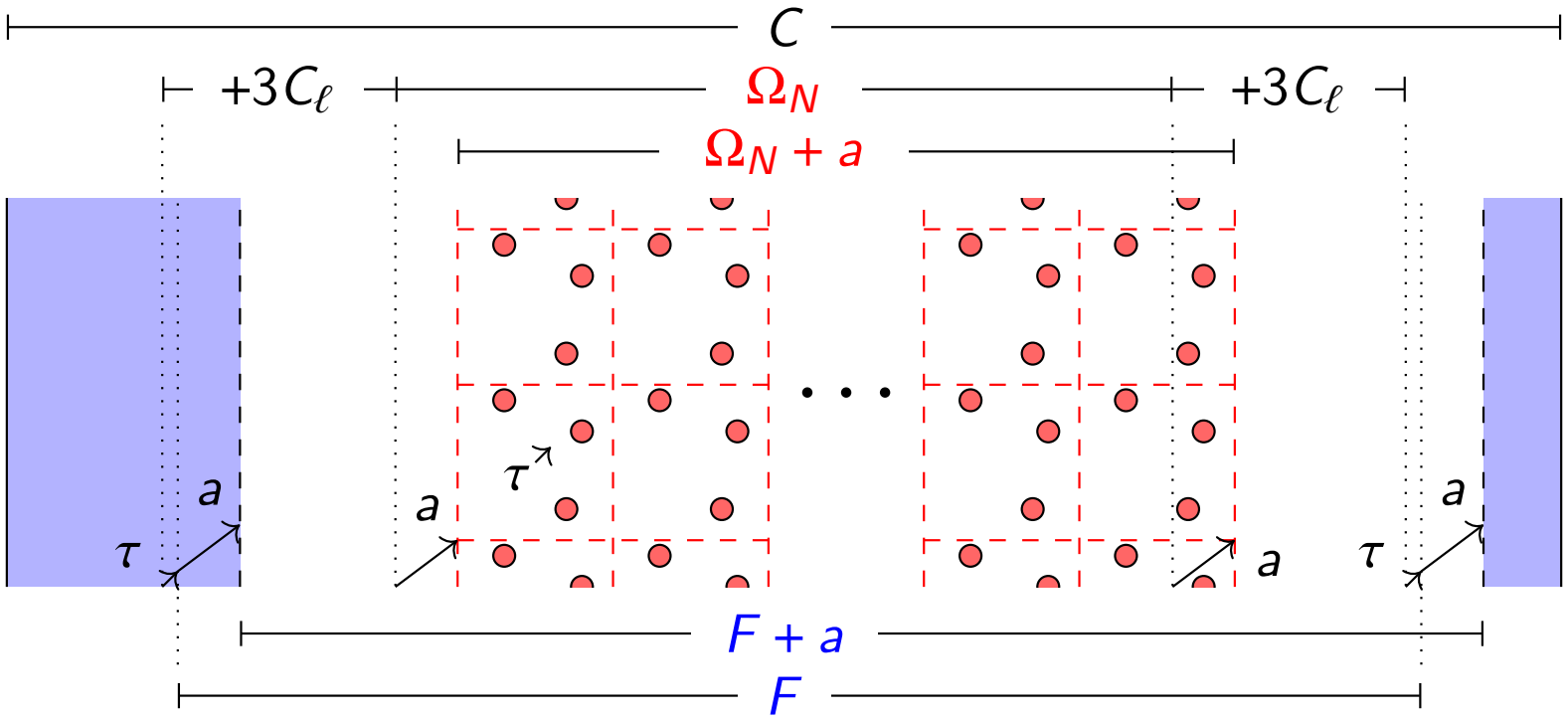
Lewin–Lieb–Seiringer gave better trial state with $\rho_{\mathbb{P}} \neq \mathbb{1}_{\Omega_N}$. Resolves problem in dimension $d = 3$. Technical issue in $d = 2$.

We need trial state $\mathbb{P}$ with density $\mathbb{1}_{\Omega_N}$.

Ideas of proof

Trial state

$$\mathbb{P} = \frac{1}{\ell^2}\int_{C_\ell} \bigotimes_{\substack{j=1,\dots,n\\ k\in\mathbb{Z}^2\\ |k_1|,|k_2|\le K}} \delta_{x_j+\ell k+a} \otimes \left(\frac{\beta - \mathbb{1}_{F+a}}{M}\right)^{\otimes M} \mathrm{d}a, \qquad \beta \approx \mathbb{1}_C, \qquad \rho_{\mathbb{P}} = \mathbb{1}_C.$$



Periodic configurations

In dimension d. Riesz interaction $V_s(x) = \mathrm{sgn}(s)|x|^{-s}$ instead of Coulomb $|x|^{2-d}$.

Theorem (L 2021)

Let $d-4 < s < d$. Let $\mathcal{L} \subset \mathbb{R}^d$ be a lattice. Then the Jellium energy of the lattice configuration is

$$e^{\mathcal{L}}_{\mathrm{Jel},s} = \begin{cases} \zeta_{\mathcal{L}}(s) & \text{if } s > 0, \\ \zeta'_{\mathcal{L}}(0) & \text{if } s = 0, \\ -\zeta_{\mathcal{L}}(s) & \text{if } s < 0. \end{cases}$$

Lattice zeta-function $\zeta_{\mathcal{L}}$ for a lattice $\mathcal{L} \subset \mathbb{R}^d$ is (analytic continuation of)

$$\zeta_{\mathcal{L}}(s) = \frac{1}{2} \sum_{x \in \mathcal{L} \setminus 0} \frac{1}{|x|^s}, \qquad \mathrm{Re}(s) > d.$$

Compute for triangular lattice. Gives $-0.66056\ldots \geq e_{\mathrm{Jel},d=2} \geq -0.66118\ldots$

Main references

- C. Cotar and M. Petrache. "Equality of the Jellium and uniform electron gas next-order asymptotic terms for Coulomb and Riesz potentials". 2019. arXiv: 1707.07664 [math-ph].
- A. B. Lauritsen. "Floating Wigner crystal and periodic Jellium configurations". *J. Math. Phys.* (forthcoming). arXiv: 2103.07975 [math-ph].
- M. Lewin, E. H. Lieb, and R. Seiringer. "Floating Wigner crystal with no boundary charge fluctuations". *Phys. Rev. B* **100**.3 (2019), p. 035127.
- M. Lewin, E. H. Lieb, and R. Seiringer. "Statistical mechanics of the uniform electron gas". *J. Ec. Polytech.–Math.* **5** (2018), pp. 79–116.